

Practical 14 - Models for rates

Statistical Models for Discrete Outcomes

London School of Hygiene & Tropical Medicine

SOLUTIONS

Datasets required: `rubber.csv`

The purpose of this exercise is to fit and interpret some Poisson regression models for data from a cohort study of men from the rubber manufacturing industry.

In one rubber factory isolated conditions were maintained for the manufacturing process, while in a second workers were exposed to more dirt and fumes. The aim is to investigate differences in death rates between men who worked in the two factories, making appropriate allowances for age.

The dataset (`rubber.csv`) consists of four variables for groups defined by factory and age category:

Variable	Description
<code>agegrp</code>	Age group: 1 = 50–59, 2 = 60–69, 3 = 70–79, 4 = 80–89
<code>factory</code>	Factory (1 or 2)
<code>deaths</code>	Number of deaths
<code>pyrs</code>	Number of man-years of exposure

Aims

The aim of this session is to address the following questions about these data:

1. Is there a difference between deaths rates in the two factories?
2. Is death rate associated with age?
3. What is the best way to model age (categorical or continuous)?

Analyses

Question 1

Calculate the total number of deaths and total person years of observation. What is the overall rate of death amongst these men? What is the log death rate?

SOLUTIONS

```
. tabstat deaths pyrs, s(sum)
```

stats	deaths	pyrs
-----+-----		
sum	160	19345

Overall deaths = 160

Total follow-up time = 19345 years Overall death rate = $160/19345 = 0.00827$ deaths per year
Log death rate = $\log(0.00827) = -4.795$ The death rate is better expressed as 8.27 deaths per 1000 years of follow-up.

Question 2

Calculate the death rate and log death rate in each factory.

SOLUTIONS

```
. tabstat deaths pyrs, s(sum) by(factory)
```

Summary statistics: sum

by categories of: factory

factory	deaths	pyrs
-----+-----		
1	72	9774
2	88	9571
-----+-----		
Total	160	19345

In factory 1: Death rate = $72/9774 = 0.00737$ deaths per year (7.37 deaths per 1000 years of follow-up). Log death rate = -4.941 .

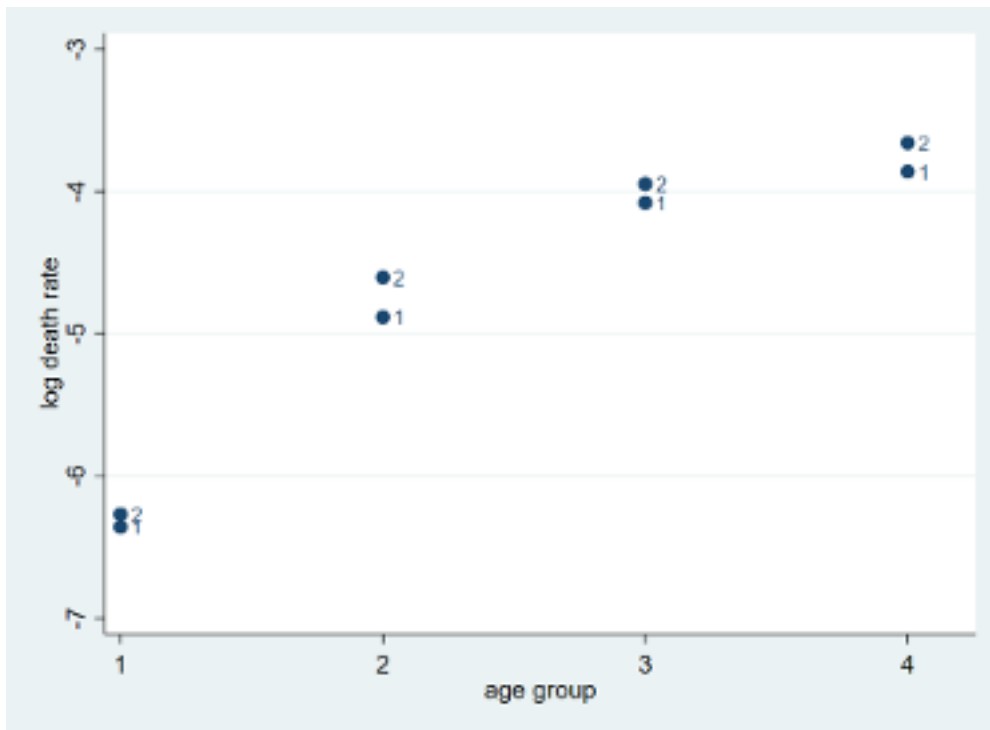
In factory 2: Death rate = $88/9571 = 0.00919$ deaths per year (9.19 deaths per 1000 years of follow-up). Log death rate = -4.689 .

Question 3

Generate a variable for the log death rates in each row, and plot these against age group, labelled according to factory (hint: use the option `mlabel([varname])` with the `twoway` command). What does this plot suggest?

What are your initial conclusions from these descriptions of the data?

SOLUTIONS



From this plot we can see that rates increase with age in both factories, and that age-specific rates are higher in factory 2 for all age groups. The age trend shows sign of flattening off at the oldest age category, suggesting some non-linearity in that relationship.

Question 4

Write down algebraically a model that can be used to investigate the relationship between death rate and age group (treated as a categorical variable, with age group 50-59 taken as the baseline category).

Take a moment to check with your colleagues that you have specified the models in the same way.

SOLUTIONS

Let Y_i be the number of deaths in group i ($i = 1, \dots, 8$) with exposure time t_i years. We assume that the Y_i are independent, with

$$Y_i \sim \text{Po}(\mu_i),$$

where

$$\mu_i = \lambda_i t_i.$$

Here λ_i is the underlying mortality rate in group i .

The link function and linear predictor are given by

$$\eta_i = \log(\mu_i) = \log(t_i) + \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}.$$

The covariates are defined as follows:

$$x_{i1} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 60--69} \\ 1 & \text{if the } i\text{th group is aged 60--69} \end{cases}$$

$$x_{i2} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 70--79} \\ 1 & \text{if the } i\text{th group is aged 70--79} \end{cases}$$

$$x_{i3} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 80--89} \\ 1 & \text{if the } i\text{th group is aged 80--89} \end{cases}$$

Question 5

Using Stata:

- (a) obtain the maximum likelihood estimates of the parameters in your model. How strong is the evidence for an age effect?

SOLUTIONS

```
. glm deaths, family(poisson) offset(lpyrs) link(log) nolog
```

Generalized linear models		Number of obs	=	8
Optimization	: ML	Residual df	=	7
		Scale parameter	=	1
Deviance	= 103.8826147	(1/df) Deviance	=	14.84037
Pearson	= 103.4540571	(1/df) Pearson	=	14.77915

Variance function: $V(u) = u$	[Poisson]
Link function : $g(u) = \ln(u)$	[Log]

	AIC	=	17.84067
Log likelihood = -70.3626661	BIC	=	89.32652

		OIM					
	deaths	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
	_cons	-4.795015	.0790569	-60.65	0.000	-4.949964	-4.640067
	lpyrs	1	(offset)				

```
. est store A
```

```
. glm deaths i.agegrp, family(poisson) offset(lpyrs) link(log)
```

Generalized linear models		Number of obs	=	8
Optimization	: ML	Residual df	=	4
		Scale parameter	=	1
Deviance	= 1.710778974	(1/df) Deviance	=	.4276947
Pearson	= 1.704679988	(1/df) Pearson	=	.42617

Variance function: $V(u) = u$	[Poisson]
Link function : $g(u) = \ln(u)$	[Log]

```

Log likelihood      = -19.27674826
AIC                 = 5.819187
BIC                 = -6.606987

```

		OIM					
deaths		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
agegrp							
60-69		1.582834	.2950484	5.36	0.000	1.004549	2.161118
70-79		2.302963	.2946408	7.82	0.000	1.725477	2.880448
80-89		2.554674	.3609046	7.08	0.000	1.847314	3.262034
_cons		-6.315874	.2672612	-23.63	0.000	-6.839697	-5.792052
lpyrs		1	(offset)				

```
. est store B
```

```
. lrtest A B
```

```

Likelihood-ratio test      LR chi2(3) = 102.17
(Assumption: A nested in B) Prob > chi2 = 0.0000

```

The test statistic here is the difference in the log likelihoods multiplied by minus 2 ($102.17 = -2 \times (-70.363 - (-19.277))$). This test gives overwhelming evidence for an effect of age.

(b) calculate the estimated rate ratios comparing age groups:

- i) 60-69 with 50-59
- ii) 80-89 with 70-79

Calculate 95% CI's for these estimates, using `lincom` (or by re-parameterising the model) for ii).

SOLUTIONS

To obtain rate ratios (and 95% confidence intervals) one approach is to use the `eform` option with the `glm` command.

```
. glm deaths i.agegrp, family(poisson) offset(lpyrs) link(log) eform nolog
```

```

Generalized linear models      Number of obs   =          8
Optimization      : ML        Residual df      =          4

```

```

Deviance          = 1.710778974
Pearson           = 1.704679988

Variance function: V(u) = u
Link function      : g(u) = ln(u)

Log likelihood     = -19.27674826

Scale parameter = 1
(1/df) Deviance = .4276947
(1/df) Pearson  = .42617

[Poisson]
[Log]

AIC = 5.819187
BIC = -6.606987

```

		OIM					
deaths		IRR	Std. Err.	z	P> z	[95% Conf. Interval]	
agegrp							
60-69		4.868732	1.436512	5.36	0.000	2.730676	8.680836
70-79		10.00378	2.94752	7.82	0.000	5.6152	17.82225
80-89		12.86711	4.643798	7.08	0.000	6.342762	26.10258
_cons		.0018074	.000483	-23.63	0.000	.0010704	.0030517
lpyrs		1	(offset)				

Note: _cons estimates baseline incidence rate.

```
. lincom 4.agegrp - 3.agegrp , eform
```

```
( 1) - [deaths]3.agegrp + [deaths]4.agegrp = 0
```

deaths		exp(b)	Std. Err.	z	P> z	[95% Conf. Interval]	
(1)		1.286225	.3503828	0.92	0.355	.7541188	2.193786

- i. The rate ratio comparing 60-69 with 50-59 year olds is 4.87(2.73, 8.68).
- ii. The rate ratio comparing 80-89 with 70-79 year olds is 1.29 (0.75, 2.19). In the Stata output above this is obtained using `lincom` with `eform`.

We can use the above `lincom` command (with `eform`) because the estimated rate ratio comparing 80-89 with 70-79 year olds (1.286225) is equal to the ratio of the estimated rate ratio comparing 80-89 with 50-59 year olds (12.86711) to the estimated rate ratio comparing 70-99 with 50-59 year olds (10.00378), i.e. $1.286225 = 12.86711/10.00378$.

See the do file for an alternative method for obtaining this rate ratio by re-parameterising.

Question 6

Now include both age group and factory effects in an additive linear predictor. What is the evidence for a difference in death rates between the factories (conditional on age group)? Calculate the estimate (and 95% CI) of the rate ratio comparing the two factories, conditional on age group.

SOLUTIONS

```
. glm deaths i.agegrp i.factory, family(poisson) offset(lpyrs) eform nolog
```

Generalized linear models		Number of obs	=	8
Optimization	: ML	Residual df	=	3
		Scale parameter	=	1
Deviance	= .2156769522	(1/df) Deviance	=	.0718923
Pearson	= .2154602966	(1/df) Pearson	=	.0718201

Variance function:	V(u) = u	[Poisson]
Link function	: g(u) = ln(u)	[Log]

	AIC	=	5.882299
Log likelihood = -18.52919725	BIC	=	-6.022648

	deaths	IRR	OIM Std. Err.	z	P> z	[95% Conf. Interval]	

	agegrp						
	60-69	4.839413	1.42806	5.34	0.000	2.714014	8.629257
	70-79	9.949875	2.931967	7.80	0.000	5.584585	17.72737
	80-89	12.86461	4.642896	7.08	0.000	6.341529	26.09751
	2.factory	1.213953	.1929612	1.22	0.223	.8889986	1.657688
	_cons	.0016398	.0004592	-22.90	0.000	.0009471	.0028389
	lpyrs	1	(offset)				

Note: _cons estimates baseline incidence rate.

```
. est store C
```

```
. lrtest B C
```

Likelihood-ratio test	LR chi2(1)	=	1.50
-----------------------	------------	---	------

(Assumption: B nested in C)

Prob > chi2 = 0.2214

The test statistic here is the difference in the log likelihoods multiplied by minus 2 ($1.50 = -2 \times (-19.277 - (-18.529))$). Note the similarity between the p-values from the Wald test of the factory effect ($z = 1.22$, hence $\chi_1^2 = 1.49$, $p = 0.223$) and the likelihood ratio test given here. These tests are asymptotically equivalent.

Conditional on age group, the death rate for factory 2 is 21% higher than that for factory 1 is (rate ratio = 1.21, 95% confidence interval (0.89, 1.66)), but this is not statistically significant.

Question 7

Now fit a model that includes the interaction between age group and factory.

- (a) What term can be used to describe this model (Hint: look at its deviance)?
- (b) What is the evidence for an interaction? How do you interpret this for these data?
- (c) Write down algebraically the linear predictor for this model, and use this to help you calculate the estimated rates of death in
 - i) factory 1, age 70-79
 - ii) factory 2, age 50-59
 - iii) factory 2, age 60-69
- (d) Check your answers against your original data

Take a moment to check with your colleagues that you have specified the models in the same way.

SOLUTIONS

```
. glm deaths i.agegrp##i.factory, family(poisson) offset(lpyrs) eform nolog
```

Generalized linear models	Number of obs	=	8
Optimization : ML	Residual df	=	0
	Scale parameter	=	1
Deviance = 4.21692e-15	(1/df) Deviance	=	.
Pearson = 1.92327e-18	(1/df) Pearson	=	.
Variance function: V(u) = u	[Poisson]		
Link function : g(u) = ln(u)	[Log]		
	AIC	=	6.60534
Log likelihood = -18.42135877	BIC	=	4.22e-15

deaths	IRR	OIM Std. Err.	z	P> z	[95% Conf. Interval]	
agegrp						
60-69	4.369123	1.853117	3.48	0.001	1.902682	10.03281
70-79	9.755608	4.094921	5.43	0.000	4.285111	22.2099
80-89	12.13348	6.279673	4.82	0.000	4.39994	33.45986
2.factory	1.092948	.5842053	0.17	0.868	.3833656	3.115917
agegrp#factory						
60-69#2	1.209462	.7153006	0.32	0.748	.379467	3.854875
70-79#2	1.043376	.6151664	0.07	0.943	.328533	3.31362
80-89#2	1.120495	.8094177	0.16	0.875	.2719716	4.616328
_cons	.0017305	.0006541	-16.83	0.000	.000825	.00363
lpyrs	1	(offset)				

Note: _cons estimates baseline incidence rate.

. est store SAT

. lrtest C SAT

Likelihood-ratio test	LR chi2(3) =	0.22
(Assumption: C nested in SAT)	Prob > chi2 =	0.9750

- (a) This is a saturated model, because the number of parameters is the same as the number of (grouped) observations. Because of this the deviance for this model is 0.
- (b) The likelihood ratio test comparing the fit of the saturated model (which includes the interaction term) with the model in part 6 (which does not) provides no evidence of an age-by-factory interaction ($p = 0.975$). Note that the test statistic here (0.22) is identical to the deviance for the model without the interaction in part 6 so strictly we did not need to fit this model in order to test for evidence of an interaction.
- (c) The linear predictor is given by

$$\eta_i = \log(t_i) + \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3} + \beta_4 z_i + \beta_5 (x_{i1} z_i) + \beta_6 (x_{i2} z_i) + \beta_7 (x_{i3} z_i).$$

The covariates are defined as follows:

$$x_{i1} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 60--69} \\ 1 & \text{if the } i\text{th group is aged 60--69} \end{cases}$$

$$x_{i2} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 70--79} \\ 1 & \text{if the } i\text{th group is aged 70--79} \end{cases}$$

$$x_{i3} = \begin{cases} 0 & \text{if the } i\text{th group is not aged 80--89} \\ 1 & \text{if the } i\text{th group is aged 80--89} \end{cases}$$

$$z_i = \begin{cases} 0 & \text{if the } i\text{th group is from factory 1} \\ 1 & \text{if the } i\text{th group is from factory 2} \end{cases}$$

To calculate estimated death rates by factory and age-group using the above specification refit the saturated model without using the `eform` option.

```
. glm deaths i.agegrp##i.factory, family(poisson) offset(lpyrs) link(log) nolog
```

< output omitted >

		OIM					
deaths		Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
agegrp							
60-69		1.474562	.4241393	3.48	0.001	.6432646	2.30586
70-79		2.277842	.4197505	5.43	0.000	1.455146	3.100538
80-89		2.495969	.5175492	4.82	0.000	1.481591	3.510346
2.factory		.0888786	.5345225	0.17	0.868	-.9587662	1.136523
agegrp#factory							
60-69#2		.1901755	.5914206	0.32	0.748	-.9689877	1.349339
70-79#2		.0424616	.5895923	0.07	0.943	-1.113118	1.198041
80-89#2		.1137709	.7223748	0.16	0.875	-1.302058	1.5296
_cons		-6.359327	.3779645	-16.83	0.000	-7.100123	-5.61853
lpyrs		1	(offset)				

For men in factory 1 aged 70–79, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_2) = \exp(-6.359 + 2.278) = 16.9 \text{ per 1000 person-years.}$$

For men in factory 2 aged 50–59, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_4) = \exp(-6.359 + 0.089) = 1.89 \text{ per 1000 person-years.}$$

For men in factory 2 aged 60–69, the estimated rate of death is

$$\exp(\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_4 + \hat{\beta}_5) = \exp(-6.359 + 1.475 + 0.089 + 0.190) = 10.0 \text{ per 1000 person-years.}$$

(d) These answers agree with the rates in the raw data.

Question 8

Now consider the age group variable as continuous. Fit a model with age group and factory as covariates and consider its deviance.

What do you conclude about the overall fit of the model with continuous age and factory as additive effects? How might the fit be improved?

SOLUTIONS

The fitted model with age group as a continuous predictor is as follows.

```
. glm deaths agegrp i.factory, family(poisson) offset(lpyrs) eform nolog
```

Generalized linear models		Number of obs	=	8
Optimization	: ML	Residual df	=	5
		Scale parameter	=	1
Deviance	= 12.71207904	(1/df) Deviance	=	2.542416
Pearson	= 11.67493152	(1/df) Pearson	=	2.334986
Variance function:	V(u) = u			[Poisson]
Link function	: g(u) = ln(u)			[Log]
		AIC	=	6.94435
Log likelihood	= -24.77739829	BIC	=	2.314871

		OIM					
deaths		IRR	Std. Err.	z	P> z	[95% Conf. Interval]	
agegrp		2.23989	.188639	9.58	0.000	1.899067	2.64188
2.factory		1.231652	.1957225	1.31	0.190	.9020346	1.681717
_cons		.0012735	.0003103	-27.36	0.000	.00079	.0020531
lpyrs		1	(offset)				

Note: _cons estimates baseline incidence rate.

```
. est store D
```

```
. lrtest D SAT
```

Likelihood-ratio test	LR chi2(5) =	12.71
(Assumption: D nested in SAT)	Prob > chi2 =	0.0262

Relating the deviance to its degrees of freedom (5) provides some evidence ($p = 0.0262$) of a lack of fit. It is probably best to revert to models which treat age as categorical.

Alternatively, one could try adding a quadratic term, but the coefficients in the categorical model are easier to interpret. Given that there are only 3 age-parameters in this model (only one less than in the quadratic model) the categorical model is probably the best choice.

Question 9

Now consider your analysis as a whole. What are your epidemiological conclusions from this analysis? Working together with one or more colleagues, write a short paragraph to summarise your findings concerning the comparison of the two factories.

SOLUTIONS

In conclusion there is strong evidence ($p < 0.001$) that death rates increase with age. The rate in 60-69 year olds was 4.8 (95% CI 2.7 to 8.6) times that in 50-59 year olds. Thereafter the rate increased in each decade but the relative decade-on-decade increases were smaller.

Death rates were 21% higher (95% CI 11% lower to 66% higher) in the second factory than in the first conditional on age group, but this difference was not statistically significant ($p = 0.22$). There was no evidence that the rate ratio comparing the two factories differed according to age.